Total No. of Questions : 8]

SEAT No. :

PE2311

[Total No. of Pages : 3

[6584]-220

B.E. (Mechanical Engineering)

AUTOMOBILE DESIGN

(2019 Pattern) (Semester - VII) (402044A) (Elective - III)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of Non-programmable scientific calculators is allowed.***
5) ***Assume suitable data, if necessary.***

Q1) a) Describe the working of power steering unit with neat diagram. **[9]**

b) A propeller shaft is required to transmit 45 kW power at 500 rpm. It is a hollow shaft, having an inside diameter 0.6 times of outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm^2. Calculate the inside and outside diameters of the shaft. **[9]**

OR

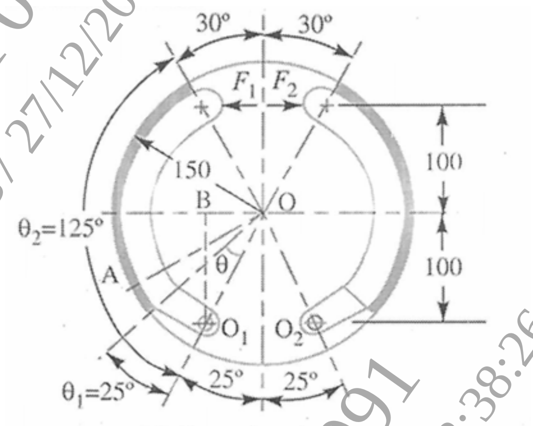
Q2) a) A pair of straight bevel gears is mounted on shafts, which are intersecting at right angles. The number of teeth on the pinion and gear are 21 and 28 respectively. The pressure angle is 20°. The pinion shaft is connected to an electric motor developing 5 kW rated power at 1440 rpm. The service factor can be taken as 1.5. The pinion and the gear are made of steel (Sut=750 N/mm^2) and heat-treated to a surface hardness of 380 BHN. The gears are machined by a manufacturing process, which limits the error between the meshing teeth to 10 mm. The module and face width are 4 mm and 20 mm respectively. Determine the factor of safety against bending failure. **[10]**

b) Difference between live axle and dead axle. **[8]**

P.T.O.

Q3) a) Differentiate between conventional tyre and tubeless tyre. **[7]**

b) Determine the braking torque and the magnitude of the forces F_1 and F_2. The figure shows the arrangement of two brake shoes which act on the internal surface of a cylindrical brake drum. The braking force F_1 and F_2 are applied as shown and each shoe pivots on its fulcrum O_1 and O_2. The width of the brake lining is 35 mm. The intensity of pressure at any point A is 0.4 sin θ N/mm², where θ is measured as shown from either pivot. The coefficient of friction is 0.4. **[10]**



All dimensions in mm.

OR

Q4) a) Explain with neat sketch construction and working of hydraulic brake system. State its applications. **[9]**

b) A circular road course track has a radius of 500 m and is banked to 10°. If the coefficient of friction between the road and tyre is 0.25. Compute (i) the maximum speed to avoid slipping and (ii) the optimum speed to avoid wear and tear on the tyres. **[8]**

Q5) a) It is required to design a helical compression spring subjected to a maximum force of 1250 N. The deflection of the spring corresponding to the maximum force should be approximately 30 mm. The spring index can be taken as 6. The spring is made of patented and cold-drawn steel wire. The ultimate tensile strength and modulus of rigidity of the spring material are 1090 and 81370 N/mm² respectively. The permissible shear stress for the spring wire should be taken as 50% of the ultimate tensile strength. Design the spring and calculate: (i) wire diameter; (ii) mean coil diameter; (iii) number of active coils; (iv) total number of coils. **[10]**

b) What is independent suspension system? Explain any one type and their advantages. **[8]**

OR

Q6) a) Explain with neat sketch multi link suspension. **[8]**

b) A semi-elliptic leaf spring used for automobile suspension consists of three extra full- length leaves and 15 graduated-length leaves, including the master leaf. The centre-to- centre distance between two eyes of the spring is 1 m. The maximum force that can act on the spring is 75 kN. For each leaf, the ratio of width to thickness is 9:1. The modulus of elasticity of the leaf material is 207000 N/mm^2 . The leaves are pre-stressed in such a way that when the force is maximum, the stresses induced in all leaves are same and equal to 450 N/mm^2. Determine (i) the width and thickness of the leaves; (ii) the initial nip; and (iii) the initial pre-load required to close the gap C between extra fulllength leaves and graduated-length leaves. **[10]**

Q7) a) Discuss standard practices used in vehicle packaging. **[8]**

b) Explain steps in driver package development. **[9]**

OR

Q8) a) Write a note on applications of biomechanics in vehicle. **[8]**

b) What is anthropometry? Explain use of anthropometry in designing vehicles. **[9]**

✪ ✪ ✪